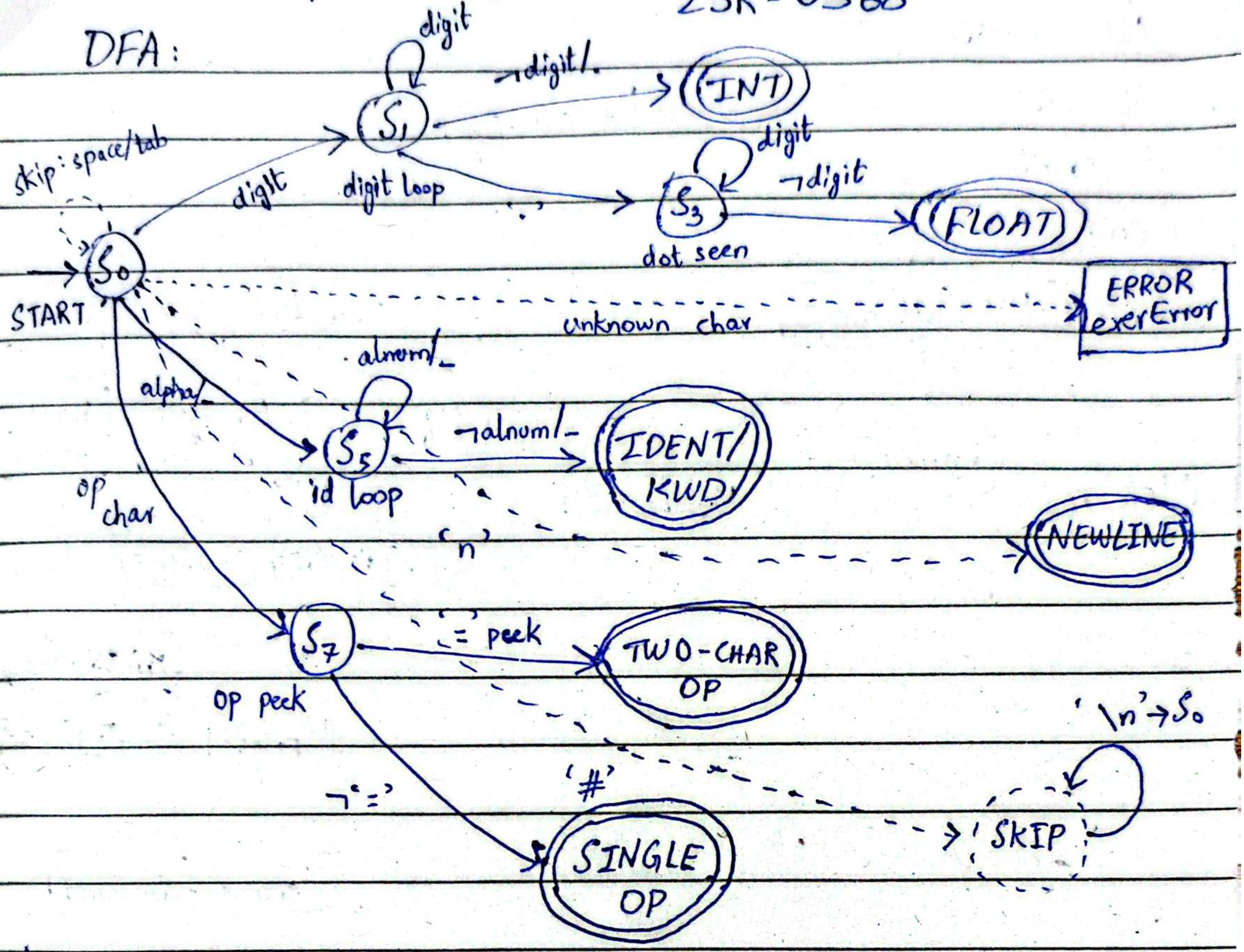
Retro Calc

23K-2001
23K-0532
23K-0568

Lexical Analyzer Design:

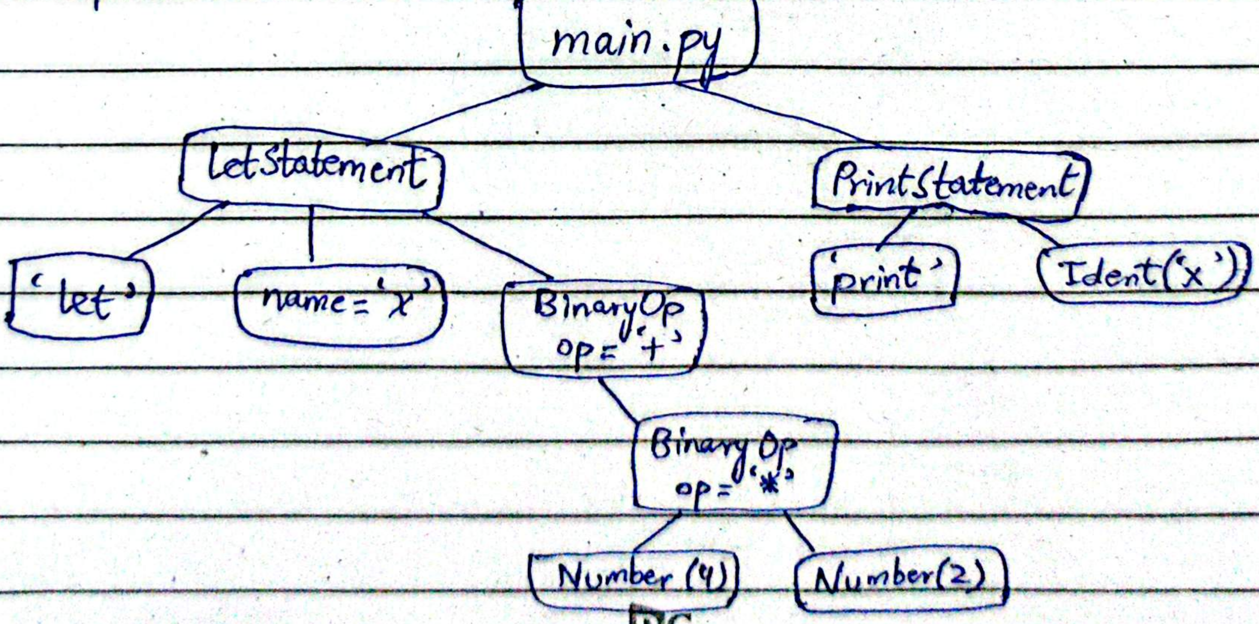
DFA:



Parser Design:

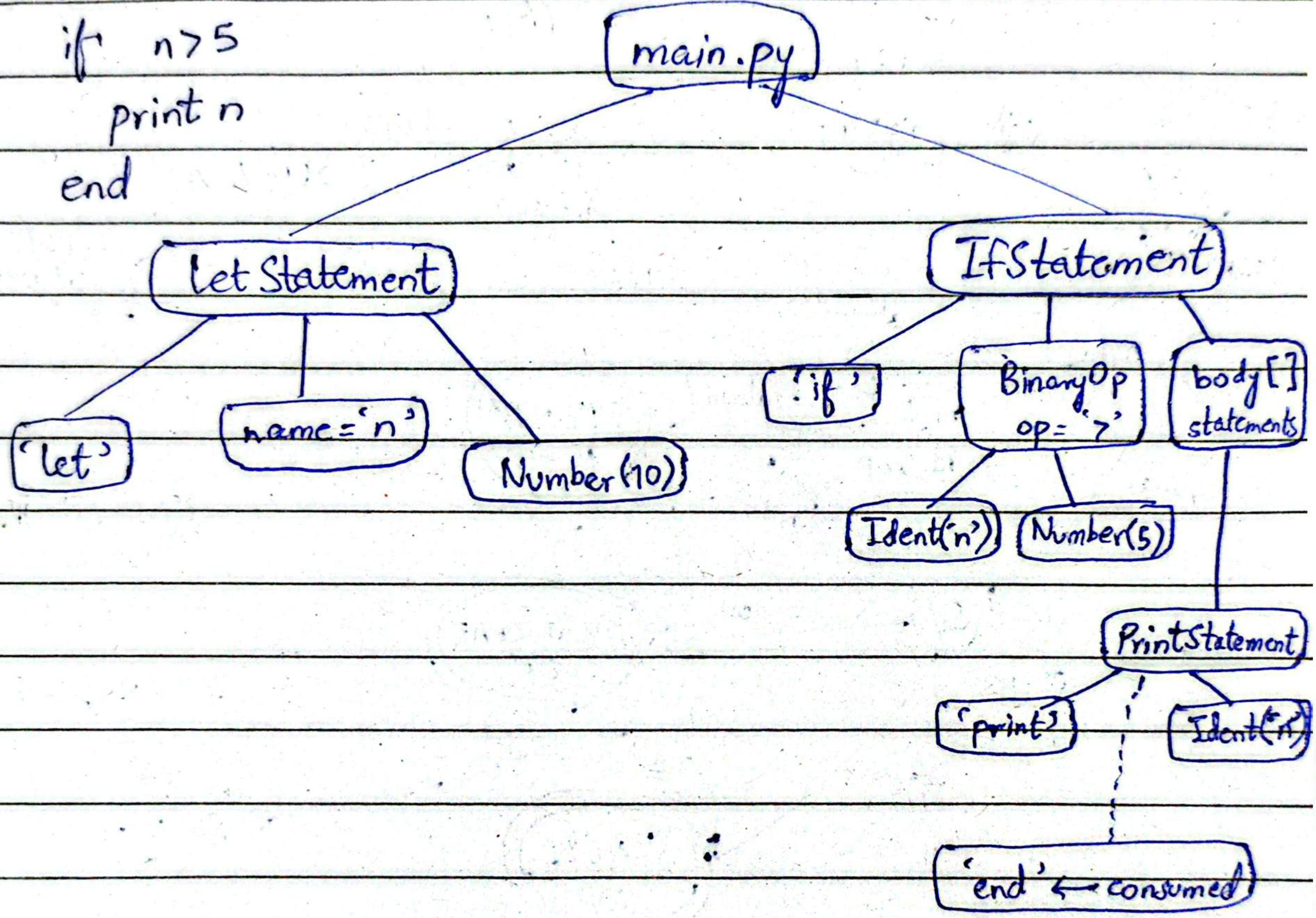
let x = 3 + 4 * 2

print x



Date

```
let n=10
if n>5
  print n
end
```



Signature

No.

Date

Semantic Analyzer Design:

Symbol table - Program 1

```
let x = 3+4*2
print x
```

variable	type	DECLARED line	Assigned From	Used	Status
x	int	1	BinaryOp(+): int+int → int	line 2	OK

Symbol table - Program 2

```
let n = 10
if n>5
  print n
end
```

variable	type	DECLARED line	Assigned From	Used	Status
n	int	1	Numberliteral(10) → int	Line 2,3	OK

Symbol table - Error case (undefined variable)

```
let x = 10
print y  # y was never declared!
```

variable	type	DECLARED line	Assigned From	Used	Status
x	int	1	NumberLiteral (10) → int	X never read	Warning
y	-	-	not in declared_vars	X ref at L2	Error

Signature

RC

No.